Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is a common application of sequence data in natural language processing? 1 / 1 point

- Database indexing
- Language modeling and text generation
- Image classification
- Data encryption

Nice work

Correct! Sequence data is used in NLP for tasks like language modeling and text generation.

2. What is a sequence in the context of data representation? 1 / 1 point

- A random assortment of data points.
- A set of unordered categorical items.
- A set of ordered categorical items.
- A numerical vector representing quantities.

Nice work

Correct. A sequence involves a set of categorical items arranged in a specific order.

3. Which of the following is an example of a sequence? 1 / 1 point

- A set of unordered survey responses.
- A collection of books in a library.
- A list of ingredients for a recipe.
- A DNA sequence of base pairs.

Nice work

Correct! A DNA sequence is a classic example where the order of base pairs is crucial.

4. Which of the following describes one way that itemset representation differs from sequence representation? 1 / 1 point

- Itemset representation is used only for numerical data, while sequence representation is for categorical data.
- Itemset representation considers the order of items, while sequence representation does not.
- Itemset representation is a type of sequence representation.
- Itemset representation ignores repeated occurrences of items, while sequence representation does not.

Nice work

Correct! Itemset representation ignores both order and repeated occurrences, unlike sequence representation.

5. In what way is a curriculum path represented differently as a sequence versus a time series? 1 / 1 point

- As a time series, only the order of courses is important.
- As a sequence, both the order and specific terms are important.
- As a sequence, the order of courses is important, but specific terms are not.
- As a time series, the order of courses can be changed.

Nice work

Correct! In a sequence, only the order matters, not the specific timing.

6. What is an example of a real-world application of sequence similarity? 1 / 1 point

- Encrypting data for security purposes.
- Calculating the average of numerical data.
- Spelling correction in search engines.
- Sorting items in a database.

Nice work

Correct! Spelling correction uses sequence similarity to suggest correct spellings based on input queries.

7. What is the computational complexity of calculating the Levenshtein edit distance between two sequences? 1 / 1 point

- O(log n)
- O(n^2)
- O(n)
- O(m*n)

Nice work

Correct! The computational complexity is O(m*n), where m and n are the lengths of the two sequences.

8. How does the Damerau-Levenshtein distance differ from the standard Levenshtein distance? 1 / 1 point

- It only considers insertions and deletions.
- It does not allow substitutions.
- It includes transposition as a single edit operation.
- It is used exclusively for DNA sequences.

Nice work

Correct! The Damerau-Levenshtein distance includes transposition as a single edit operation, unlike the standard Levenshtein distance.

9. What is the main advantage of using shingling for sequence similarity? 1 / 1 point

- It allows for efficient similarity measurement using overlapping ngrams.
- It has a time complexity of O(m×n).
- It is more accurate than edit distance for all sequence comparisons.
- It preserves the global order of sequences.

Nice work

Correct! Shingling uses overlapping ngrams to efficiently measure sequence similarity.

10. Why is edit distance not suitable for comparing long sequences? 1 / 1 point

- It cannot handle overlapping ngrams.
- It has a quadratic time complexity, making it inefficient.
- It is not accurate for long sequences.
- It does not provide a bounded similarity range.

Nice work

Correct! Edit distance has a quadratic time complexity (O(m×n)), making it inefficient for long sequences.

11. Which of the following best describes a trigram? 1 / 1 point

- A sequence of three consecutive items.
- A sequence of three items with gaps allowed.
- A sequence of items that includes a prefix and a suffix.
- A sequence of two consecutive items.

Nice work

Correct. A trigram is an N-gram of length 3, meaning it contains three consecutive items in a sequence.

12. Why might skip-grams be preferred over N-grams in certain NLP tasks? 1 / 1 point

- Because skip-grams are easier to compute than N-grams.
- Because skip-grams can capture non-consecutive dependencies between words.
- Because skip-grams only consider the first and last word of a sequence.
- Because skip-grams ensure all items are adjacent.

Nice work

Correct. Skip-grams are useful in NLP because they can capture meaningful relationships between words that are not right next to each other in a sentence.